

PineRE Development Log (Detailed)

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Executive Summary

Over approximately 30 hours of focused USB investigation, PineRE progressed from an unknown USB state to a fully operational USB ECM network connection capable of successful IPv4 communication.

Phase 19 - First Kernel Boot

Achievement

Successfully reached a login prompt from the custom PineRE image.

Proven Components

- Bootloader functional
- Kernel loading correctly
- Device Tree loading correctly
- Root filesystem mounting
- init process operational
- Console login available

Phase 20-21 - Hardware Investigation

Initial Problem

USB SSH did not function.

Unknowns

- OTG functionality
- USB PHY status
- UDC availability
- Gadget framework presence

Discoveries

- UDC exists
- OTG subsystem present
- Host cable detection works
- PHY detects laptop connection

Phase 22 - ConfigFS Discovery

Initially usb_gadget was missing.

Investigation revealed ConfigFS support existed but was not mounted.

Successful command:

```
mount -t configfs none /sys/kernel/config
```

Result:

- ConfigFS confirmed functional

Phase 23 - libcomposite Discovery

Loading:

modprobe libcomposite

Created:

/sys/kernel/config/usb_gadget

Result:

- Gadget framework confirmed operational

Phase 24 - First Gadget Creation

Created first USB gadget.

Initial bind attempt failed:

Invalid argument

Root cause:

- Missing configuration
- Missing functions

Result:

- Gadget framework proven functional

Phase 25 - ECM Gadget Construction

Constructed complete ECM gadget.

Laptop detected:

Product: PineRE ECM Test

Manufacturer: PineRE

Result:

- USB descriptors working
- Host enumeration working

Phase 26 - Enumeration Success

Laptop reported:

cdc_ether

and created:

usb0

Result:

- ECM gadget operational

Phase 27 - Network Bring-Up

Handheld reported:

configured

and:

high-speed

Result:

- USB network fully established

Phase 28 - IPv4 Configuration

Laptop:

10.42.0.1/24

PineRE:

10.42.0.2/24

Phase 29 - ARP Success

Neighbour table reported:

REACHABLE

Result:

- Ethernet frames successfully traversing USB

Phase 30 - Ping Success

Successful communication:

64 bytes from 10.42.0.2

Packet loss:

0%

Result:

- USB networking fully operational

Lessons Learned

- ConfigFS support existed all along
- libcomposite was required before usb_gadget appeared
- Incomplete gadgets cannot bind to the UDC
- ECM was easier to validate than SSH directly
- Incremental testing dramatically accelerated progress

Current Status

Completed:

- First boot
- USB gadget framework
- ECM networking
- IPv4
- ARP
- ICMP ping

Next:

- SSH daemon discovery
- SSH over USB

